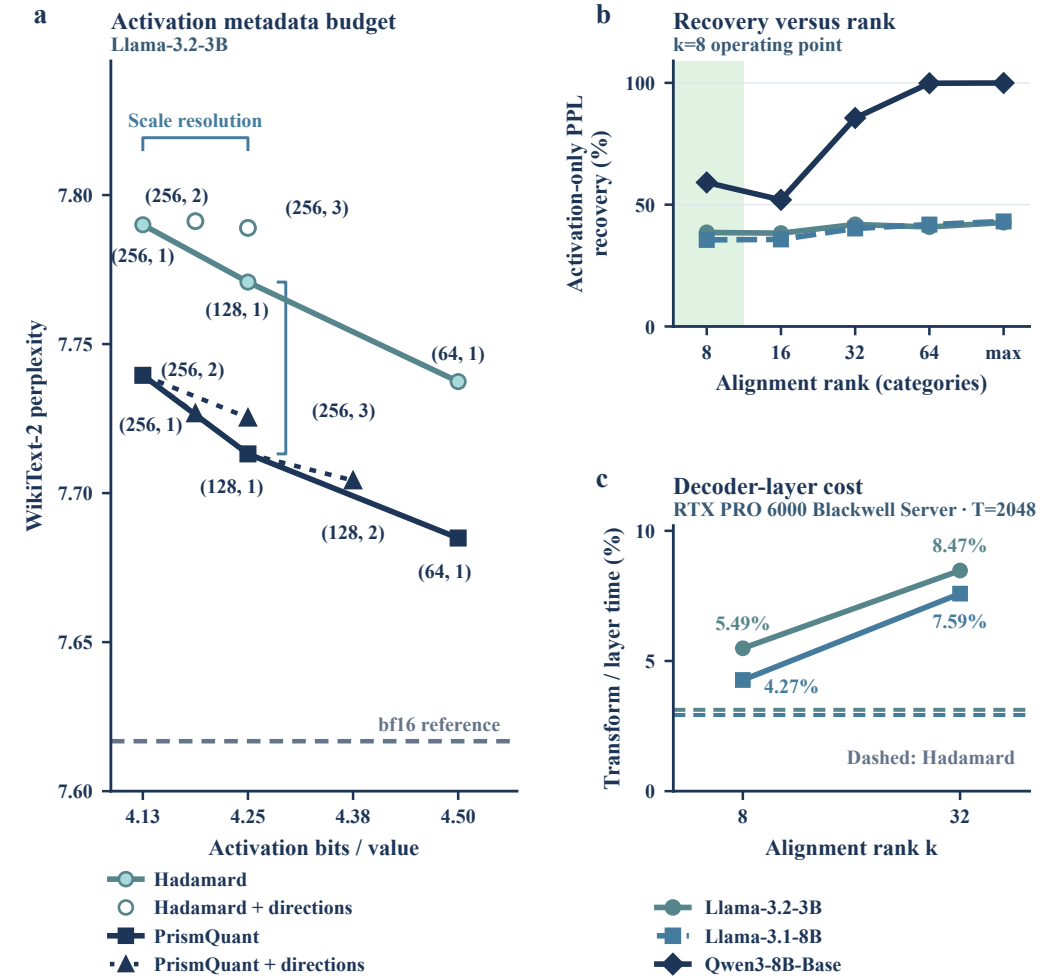
a
Activation metadata budget
Llama-3.2-3B
Scale resolution
(256, 2)
(256, 3)
(256, 1)
(128, 1)
(64, 1)
(256, 2)
(256, 3)
(256, 1)
(128, 1)
(128, 2)
(64, 1)
bf16 reference
WikiText-2 perplexity
7.80
7.75
7.70
7.65
7.60
4.13
4.25
4.38
4.50
Activation bits / value
Hadamard
Hadamard + directions
PrismQuant
PrismQuant + directions
b
Recovery versus rank
k=8 operating point
Activation-only PPL recovery (%)
100
50
0
8
16
32
64
max
Alignment rank (categories)
c
Decoder-layer cost
RTX PRO 6000 Blackwell Server · T=2048
Transform / layer time (%)
10
5
0
8.47%
5.49%
7.59%
4.27%
Dashed: Hadamard
8
32
Alignment rank k
Llama-3.2-3B
Llama-3.1-8B
Qwen3-8B-Base